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5 Unequal probability sampling

Sampling with **unequal** inclusion probabilities (unequal probability sampling):

- is of **interest in its own right**,
- provides the **basis** for formulating a **general theory of sampling**.

For orientation

In Chapters 1–4, we followed a **top-down approach**, i.e., we

- specified a **probability distribution** $\mathbb{P}$ over the set of all possible samples $\mathscr{S}$ and
- thereby defined the **inclusion probabilities**

$$\pi_i := \mathbb{P}(i \in s), \qquad i \in U.$$

In what follows, we take a **bottom-up approach**,

- in which the inclusion probabilities π_i are assumed to be **known** and
- serve as the **starting point** for constructing a sampling design with these inclusion probabilities.

Motivation

- We are interested in estimating the population total

$$t_y = \sum_{i \in U} y_i.$$

- Recall from Chapter 4 that estimators under **stratified** simple random sampling (STSI) can, under suitable conditions, be **more efficient** than under SRS.

- Under STSI, the inclusion probabilities n_h/N_h **differ across strata** $h = 1, \ldots, H$ (except under proportional allocation).

- The estimator $\widehat{t}^{\,\mathrm{strat}}$ under STSI **compensates** for the unequal inclusion probabilities across strata by using the **weights** N_h/n_h, i.e., the **inverse inclusion probabilities**:

$$\widehat{t}_y^{\,\mathrm{strat}} = \sum_{h=1}^{H} \frac{N_h}{n_h} \sum_{i \in s_h} y_i.$$

- We now **generalize** this idea as follows:
  - **all** elements $i \in U$ may have **different** inclusion probabilities π_i,
  - each observation is **weighted** by the **inverse** $1/\pi_i$.

5.1 The Horvitz–Thompson estimation principle

Definition 11 (Horvitz and Thompson, 1952).

Let p be a sampling design without replacement with **possibly unequal** inclusion probabilities $\pi_i > 0$ $(i \in U)$. The **Horvitz–Thompson** (HT) estimator of the population total t_y is

$$\widehat{t}_y^{\mathrm{HT}} = \sum_{i \in s} \frac{y_i}{\pi_i}.$$

- The estimator should more properly be called the **Narain–Horvitz–Thompson** estimator; Narain (1951) proposed it "solely on intuitive grounds" (Ajgaonkar, 1967, p. 55).
- **Alternative** representation: $\widehat{t}_y^{\mathrm{HT}} = \sum_{i \in s} w_i y_i$ with $w_i = \frac{1}{\pi_i}$

Aside (Unbiasedness of the HT estimator).

Let $\mathbb{1}_i$ denote the sample inclusion indicator $\mathbb{1}\{i \in s\}$. It is straightforward to show that the HT estimator is **unbiased** for the population total t_y for **arbitrary** inclusion probabilities $\pi_i > 0$ $(i \in U)$:

$$\mathbb{E}_p(\widehat{t}_y^{\mathrm{HT}}) = \mathbb{E}_p\left(\sum_{i\in U} \mathbb{1}_i \frac{y_i}{\pi_i}\right) = \sum_{i\in U} \mathbb{E}_p(\mathbb{1}_i)\frac{y_i}{\pi_i} = \sum_{i\in U} \pi_i \frac{y_i}{\pi_i} = \sum_{i\in U} y_i = t_y,$$

where we used $\mathbb{E}_p(\mathbb{1}_i) = \pi_i$.

- The choice of the π_i **does**, however, affect the variance of the HT estimator.
- All estimators discussed so far are **special cases** of the HT estimator (e.g., $\widehat{t}_y$ under SRS or $\widehat{t}_y^{\mathrm{strat}}$ under STSI).

Estimating the population mean

- The population mean $\bar{y}_U$ can be estimated using the HT estimator only if N is **known**:

$$\bar{y}^{\,\mathrm{HT}} = \frac{\widehat{t}_y^{\,\mathrm{HT}}}{N}.$$

- If N is **unknown**, the **Hájek estimator** (HJ) can be used (Hájek, 1971, p. 236):

$$\bar{y}^{\,\mathrm{HJ}} = \frac{\sum_{i\in s} y_i/\pi_i}{\sum_{i\in s} 1/\pi_i} = \frac{\sum_{i\in s} w_i y_i}{\sum_{i\in s} w_i} \qquad \text{with} \quad w_i = \frac{1}{\pi_i},$$

$$= \frac{1}{\widehat{N}} \sum_{i\in s} w_i y_i = \frac{\widehat{t}_y^{\,\mathrm{HT}}}{\widehat{N}} \qquad \text{with} \quad \widehat{N} = \sum_{i\in s} w_i.$$

- The estimator $\widehat{N}$ can be viewed as the HT estimator of N with $y_i \equiv 1$, i.e., $\widehat{N} = \sum_{i \in s} 1 \, w_i$.

- The **Hájek estimator** is a **ratio estimator**:

$$\bar{y}^{\,\mathrm{HJ}} = \frac{\widehat{t}_y^{\,\mathrm{HT}}}{\widehat{N}}$$

$\Rightarrow$ and is generally **biased** in finite samples, although,

$\Rightarrow$ the bias is typically small for sufficiently **large samples**.

$\Rightarrow$ is frequently used in practice even when N is **known**.

5.2 Variance estimation

Definition 12 (Second-order inclusion probabilities).

Let p be a sampling design without replacement with $\pi_i > 0$ for all $i \in U$. The probability that elements i and j are both included in the sample s is called the **second-order inclusion probability** and is defined as

$$\pi_{ij} := \mathbb{P}(i,j \in s) = \mathbb{P}(\mathbb{1}_i \mathbb{1}_j = 1) \qquad (i,j \in U).$$

- For the **special case** $i = j$, $\pi_{ii} = \pi_i$ for all $i \in U$.
- For a population of N elements, the matrix of second-order inclusion probabilities is of size $(N \times N)$. Because it is **symmetric**, it has

$$\underbrace{N}_{\pi_{ii}=\pi_i} + \underbrace{\frac{N(N-1)}{2}}_{\pi_{ij}, i<j}$$ nonredundant entries.

Definition 13 (Covariance).

Let p be a sampling design without replacement with $\pi_i > 0$ for all $i \in U$. The **covariance** Δ_{ij} between the inclusion indicators $\mathbb{1}_i = \mathbb{1}\{i \in s\}$ and $\mathbb{1}_j = \mathbb{1}\{j \in s\}$, $i, j \in U$, is defined as

$$\Delta_{ij} := \mathrm{cov}_p(\mathbb{1}_i, \mathbb{1}_j) = \pi_{ij} - \pi_i \pi_j.$$

- Depending on the design, the covariances may be **zero**, **positive**, or **negative**.
- For the variance,

$$\mathrm{var}_p(\mathbb{1}_i) = \pi_i(1 - \pi_i) \qquad \text{for all} \quad i \in U.$$

Example 18 (SRS).

Under SRS with sample size n $(0 < n < N)$, the inclusion probabilities are $\pi_i = n/N$ for all $i \in U$ and

$$\pi_{ij} = \frac{n(n-1)}{N(N-1)} = \pi_i \pi_j \frac{N(n-1)}{n(N-1)} \qquad (\text{for } i \neq j,\ i,j \in U).$$

The **covariance** between the inclusion indicators $\mathbb{1}_i$ and $\mathbb{1}_j$ is

$$\Delta_{ij} = \pi_{ij} - \pi_i \pi_j = \frac{n(n-1)}{N(N-1)} - \left(\frac{n}{N}\right)^2 = -\frac{f(1-f)}{N-1} \qquad (\text{for } i \neq j, i,j \in U)$$

with $f = n/N$. Since $0 < n < N$, we have $0 < f < 1$. Hence the covariance is strictly **negative** for all $i \neq j$.

- Let p be a sampling design without replacement with $\pi_i > 0$ for all $i \in U$. The theoretical **variance of the HT estimator** is

$$\text{var}_p(\widehat{t}^{\text{HT}}) = \sum_{i \in U} \sum_{j \in U} \Delta_{ij} \frac{y_i y_j}{\pi_i \pi_j}$$

- If $\pi_{ij} > 0$ for all $i, j \in U$, then $\text{var}_p(\widehat{t}^{\text{HT}})$ can be estimated by the unbiased **Horvitz–Thompson variance estimator** (Horvitz and Thompson, 1952):

$$\widehat{\text{var}}_p^{HT}(\widehat{t}^{\text{HT}}) = \sum_{i \in s} \sum_{j \in s} \frac{\Delta_{ij}}{\pi_{ij}} \frac{y_i y_j}{\pi_i \pi_j}.$$

- For **fixed-size sampling designs**, alternative unbiased variance estimators are available; see Supplements (Sect. 5.4B).
- **Approximations** to second-order inclusion probabilities can be useful when the exact π_{ij} are unavailable or difficult to compute; see Supplements (Sect. 5.4C).

5.3 Probability proportional to size (PPS)

Example 19 (Oracle, Population U14).

i:	1	2	3	4	5	6	7	8	9	10	11	12	13	14
y_i:	175	143	116	85	150	92	108	67	121	99	118	259	75	142
π_i:	$\frac{2}{5}$	$\frac{286}{875}$	$\frac{232}{875}$	$\frac{34}{175}$	$\frac{12}{35}$	$\frac{184}{875}$	$\frac{216}{875}$	$\frac{134}{875}$	$\frac{242}{875}$	$\frac{198}{875}$	$\frac{236}{875}$	$\frac{74}{125}$	$\frac{6}{35}$	$\frac{284}{875}$

A probability sample of size $n = 4$ is selected without replacement from U14. The **realized** sample s is

$$y_1 = 175, \quad y_2 = 143, \quad y_8 = 67, \quad y_{14} = 142$$
$$\pi_1 = \frac{2}{5}, \quad \pi_2 = \frac{286}{875}, \quad \pi_8 = \frac{134}{875}, \quad \pi_{14} = \frac{284}{875}.$$

Example (Cont.).

- For the realized sample, the HT estimator gives

$$\widehat{t}_y^{\mathrm{HT}} = \sum_{i \in s} \frac{y_i}{\pi_i} = 1'750.$$

- The estimate $\widehat{t}_y^{\mathrm{HT}}$ equals the true population total t_y.
- The variance of the estimator is zero, $\mathrm{var}_p(\widehat{t}_y^{\mathrm{HT}}) = 0$; hence there is **no sampling uncertainty**.
- In this application, the HT estimator has zero variance because the inclusion probabilities π_i were constructed by an **oracle**, assuming that all y_i $(i \in U)$ were already **known before sampling**.
- Clearly, this makes the idea of estimation **absurd**.

Example (Cont.).

- We can nevertheless learn something from this **absurd** situation.
- The first-order inclusion probabilities π_i were chosen **proportional** to the y_i:

$$\pi_i = \frac{ny_i}{\sum_{i \in U} y_i} = \frac{ny_i}{t_y} \qquad (\text{for all } i \in U),$$

so that the HT estimator becomes

$$\widehat{t}_y^{\mathrm{HT}} = \sum_{i \in s} \frac{y_i}{\pi_i} = \sum_{i \in s} \frac{y_i}{ny_i} t_y = t_y.$$

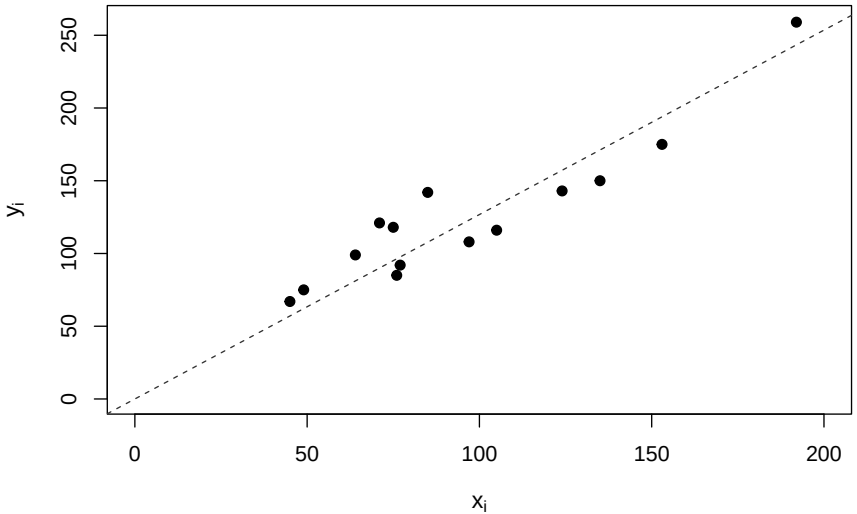
- Let x_i be an **auxiliary variable** known for **all** $i \in U$.
- For example, $x_i =$ number of employees in business i (auxiliary variable known from a register), and $y_i =$ revenue (variable of interest, unknown).

- Assume that y_i and x_i are strongly **positively correlated**.
- We consider a fixed-size sampling design. If $y_i \propto x_i$, then $\text{var}_p(\widehat{t}_y^{\text{HT}}) = 0$; see the artificial situation in Example 19.
- The first-order **inclusion probabilities** are chosen proportional to size (PPS), i.e.,

$$\pi_i = \frac{n x_i}{\sum_{i \in U} x_i} \qquad (i \in U),$$

  provided that $0 < \pi_i \leq 1$ for all $i \in U$.
- Variable x_i is called the **measure of size** (e.g., number of employees).
- If proportionality is not exact, PPS sampling can still yield **efficiency gains** over SRS when x_i is strongly related to y_i.

Example (Data from Example 19)



5.4. Supplements

5.4A Supplement: Selecting a PPS sample

- Hanif and Brewer (1980) classified **47** PPS sampling procedures without replacement; for more recent literature, see Tillé (2006).
- Following their classification, we impose the following **requirements** on PPS designs (for all $i, j \in U$):
  - **fixed** sample size n;
  - the π_i are strictly **proportional** to x_i;
  - **unbiased** variance estimation: $\pi_{ij} > 0$ for all $i \neq j$;
  - the SYG variance estimator is **nonnegative**: $\Delta_{ij} \leq 0$ for all $i \neq j$.
- The sampling procedure of Brewer (1963, 1975) satisfies these requirements (first procedure).

Sampling procedure of Brewer (1963); sample size $n = 2$

Assumption: $x_i < t_x/2$ for all $i \in U$. Then

$$\pi_i = \frac{2x_i}{t_x} < 1 \qquad (i \in U).$$

1) Select the **first element** with probability

$$p_i = \frac{c_i}{\sum_{i \in U} c_i}, \qquad c_i = \frac{x_i(t_x - x_i)}{t_x(t_x - 2x_i)} \qquad (i \in U).$$

Denote the selected element by k; it is not replaced.

2) Select the **second element** with conditional probability

$$p_{j|k} = \frac{x_j}{t_x - x_k}, \qquad j \in U \setminus \{k\}.$$

- Brewer (1963) derived a formula for the **second-order inclusion probabilities**.
- In the **general case**, i.e., when $n > 2$,
  - the prescribed PPS inclusion probabilities

$$\pi_i = \frac{nx_i}{t_x}$$

    require $x_i \leq t_x/n$ for all $i \in U$;
  - units with $\pi_i = 1$ are **certainty units** and are included in the sample with probability one;
  - after removing certainty units, the inclusion probabilities for the remaining units must be recalculated;
  - sample selection is substantially more **involved**; see Brewer (1975).

5.4B Supplement: Fixed sample size

- For a sampling design with **fixed sample size** n, $\sum_{j \in U} \Delta_{ij} = 0$ for all $i \in U$.
- Hence, the variance of the HT estimator can also be written as

$$\text{var}_p\left(\hat{t}^{\text{HT}}\right) = -\frac{1}{2} \sum_{i \in U} \sum_{j \in U} \Delta_{ij} \left(\frac{y_i}{\pi_i} - \frac{y_j}{\pi_j} \right)^2 .$$

- If $\pi_{ij} > 0$, this leads to the **Sen–Yates–Grundy** (SYG; Sen, 1953; Yates and Grundy, 1953) variance estimator

$$\widehat{\text{var}}_p^{SYG}\left(\hat{t}^{\text{HT}}\right) = -\frac{1}{2} \sum_{i \in s} \sum_{j \in s} \frac{\Delta_{ij}}{\pi_{ij}} \left(\frac{y_i}{\pi_i} - \frac{y_j}{\pi_j} \right)^2 .$$

- If $\Delta_{ij} \leq 0$ for all $i \neq j$, $i, j \in U$, the SYG estimator is **nonnegative**.
- The SYG estimator is **numerically more stable** than $\widehat{\text{var}}_p^{HT}$, particularly when the coefficient of variation of the y_i/π_i is small.

Aside (Important identities).

For sampling designs p with **fixed sample size** n,

$$\sum_{i \in U} \pi_i = n \qquad \text{and} \qquad \sum_{j \in U} \pi_{ij} = n\pi_i \quad (i \in U).$$

Hence,

$$\sum_{j \in U} \Delta_{ij} = \sum_{j \in U} (\pi_{ij} - \pi_i \pi_j) = 0 \qquad (i \in U).$$

Aside (cont.).

- In other words,

$$\text{fixed } n \implies \sum_j \pi_{ij} = n\pi_i \implies \sum_j \Delta_{ij} = 0 \implies \text{SYG representation}$$

- The above identities are useful for checking inclusion probabilities and lead to the Sen–Yates–Grundy representation of the HT variance.

5.4C Supplement: Approximation of second-order inclusion probabilities

- Brewer (2000) proposed the following **approximation** of second-order inclusion probabilities (for high-entropy) designs:

$$\pi_{ij} \approx \pi_i \pi_j \left(\frac{c_i + c_j}{2} \right) \qquad (i, j \in U,\, i \neq j),$$

  where the choice of the constants c_i, $i \in U$, is discussed below.

- Based on the above approximation, Brewer (2000) gave the following approximation to the theoretical **variance of the HT estimator**:

$$\text{var}_p^{\text{A}}\left(\widehat{t}^{\,\text{HT}}\right) = \sum_{i \in U} \pi_i (1 - c_i \pi_i) \left(\frac{y_i}{\pi_i} - \frac{t_y}{n} \right)^2.$$

- He also recommended the following **variance estimator**:

$$\widehat{\text{var}}_p^{\text{A}}\left(\widehat{t}^{\text{HT}}\right) = \sum_{i \in s} \left(\frac{1}{c_i} - \pi_i\right) \left(\frac{y_i}{\pi_i} - \frac{\widehat{t}_y^{\text{HT}}}{n}\right)^2.$$

  - The variance estimator is asymptotically unbiased for $\text{var}_p^{\text{A}}\left(\widehat{t}^{\text{HT}}\right)$.
  - $\widehat{\text{var}}_p^{\text{A}}\left(\widehat{t}^{\text{HT}}\right)$ is easy to compute.

- Brewer (2000) proposed **three variants** for choosing the **constants** c_i. Variant 1 uses:

$$c_i = \frac{n-1}{n-\pi_i} \qquad (i \in U).$$

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